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1) Matrix and Solution of System of Linear Equation

Question:

Using Gauss-elemination method solve the following system of linear equation:

$$\left.\begin{aligned} x+2y+z &= 4 \\ 2x+3y+4z &= 10 \\ -x+y+2z &= 1 \end{aligned}\right\}$$

Answer:

Given system,

$$\left.\begin{aligned} x+2y+z &= 4 \\ 2x+3y+4z &= 10 \\ -x+y+2z &= 1 \end{aligned}\right\} \quad \text{--- (1)}$$

The above system (1) is eqnivalent to $AX = B$

where, $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ -1 & 1 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and, $B = \begin{bmatrix} 4 \\ 10 \\ 1 \end{bmatrix}$

Now our aim is to reduce the augmented matrix of (1) to upper triangular matrix.

Augmented matrix of (1) is:

$$[A:B] = \left[\begin{array}{ccc:c} 1 & 2 & 1 & 4 \\ 2 & 3 & 4 & 10 \\ -1 & 1 & 2 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc:c} 1 & 2 & 1 & 4 \\ 0 & -1 & 2 & 2 \\ 0 & 3 & 3 & 5 \end{array}\right] \qquad \begin{aligned} r_2' &= r_2 - 2r_1 \\ r_3' &= r_3 + r_1 \end{aligned}$$

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$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -1 & 2 & 2 \\ 0 & 0 & 9 & 11 \end{array}\right]$$

$\therefore$ The reduce system is,

$$\left.\begin{aligned} x+2y+z &= 4 \\ -y+2z &= 2 \\ 9z &= 11 \end{aligned}\right\}$$

Now, by back substitute,

$$9z = 11$$

$$\therefore z = \frac{11}{9}$$

$$-y + 2 \times \frac{11}{9} = 2$$

$$\Rightarrow y = \frac{4}{9}$$

$$x + 2 \times \frac{4}{9} + \frac{11}{9} = 4$$

$$\Rightarrow x = 4 - \frac{8}{9} - \frac{11}{9}$$

$$\therefore x = \frac{17}{9}$$

$\therefore$ The required solution, $x = \frac{17}{9}$, $y = \frac{4}{9}$ and $z = \frac{11}{9}$

Ans

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2) Vector Space :

Question:

Show ~~that~~ whether the vector $v = (1,5,4)$ is a linear combination of the vectors $v_1 = (2,2,0)$, $v_2 = (0,1,1)$, $v_3 = (2,5,1)$. or not

Answer:

In order to show that, v is a linear combination of v_1, v_2, v_3 there must be scaler d_1, d_2, d_3 in F such that, $v = d_1 v_1 + d_2 v_2 + d_3 v_3$

$\Rightarrow (1,5,4) = d_1(2,2,0) + d_2(0,1,1) + d_3(2,5,1)$

$\Rightarrow (1,5,4) = (2d_1, 2d_1) + (d_2, d_2) + (2d_3, 5d_3, d_3)$

$\Rightarrow (1,5,4) = (2d_1 + 2d_3, 2d_1 + d_2 + 5d_3, d_2 + d_3)$

$\Rightarrow 2d_1 + 2d_3 = 1$, $2d_1 + d_2 + 5d_3 = 5$, $d_2 + d_3 = 4$

This system can be written as,

$$\begin{bmatrix} 2 & 0 & 2 \\ 2 & 1 & 5 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}$$

Let, $A = \begin{bmatrix} 2 & 0 & 2 \\ 2 & 1 & 5 \\ 0 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ and, $B = \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}$

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Solving with cramers' rule:

$$\det(A) = \begin{vmatrix} 2 & 0 & 2 \\ 2 & 1 & 5 \\ 0 & 1 & 1 \end{vmatrix} = 2(1-5) + 2(2-0)$$
$$= -4$$

$$\det(A_1) = \begin{vmatrix} 1 & 0 & 2 \\ 5 & 1 & 5 \\ 4 & 1 & 1 \end{vmatrix} = 1(1-5) + 2(5-4)$$
$$= -2$$

$$\det(A_2) = \begin{vmatrix} 2 & 1 & 2 \\ 2 & 5 & 5 \\ 0 & 4 & 1 \end{vmatrix} = 2(5-20) - 1(2-0) + 2(8-0)$$
$$= -16$$

$$\det(A_3) = \begin{vmatrix} 2 & 0 & 1 \\ 2 & 1 & 5 \\ 0 & 1 & 4 \end{vmatrix} = 2(4-5) + 1(2-0)$$
$$= 0$$

$$\therefore d_1 = \frac{\det(A_1)}{\det(A)} = \frac{-2}{-4} = \frac{1}{2}$$

$$\therefore d_2 = \frac{\det(A_2)}{\det(A)} = \frac{-16}{-4} = 4$$

$$\therefore d_3 = \frac{\det(A_3)}{\det(A)} = \frac{0}{-4} = 0$$

Hence, $v = \frac{1}{2} v_1 + 4 v_2$.

Therefore v is not a linear combination of v_1, v_2 and v_3

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3) Linear Transformation.

Question:

Show that, the mapping $T: R^2 \to R^2$ defined by $T(x,y) = (3x+y, x+2y)$ is a linear transformation.

Answer:

Let, $u_1 = (x_1, y_1)$ and $u_2 = (x_2, y_2) \in R^2$

$\therefore u_1 + u_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

Now, $T(u_1) = T(x_1, y_1) = (3x_1 + y_1, x_1 + 2y_1)$

$T(u_2) = T(x_2, y_2) = (3x_2 + y_2, x_2 + 2y_2)$

$\therefore T(u_1 + u_2) = T(x_1 + x_2, y_1 + y_2)$

$= 3(x_1 + x_2) + y_1 + y_2, \; x_1 + x_2 + 2(y_1 + y_2)$

$= 3x_1 + 3x_2 + y_1 + y_2, \; x_1 + x_2 + 2y_1 + 2y_2$

$= (3x_1 + y_1, x_1 + 2y_1) + (3x_2 + y_2, x_2 + 2y_2)$

$= T(u_1) + T(u_2)$

$\therefore T(u_1 + u_2) = T(u_1) + T(u_2)$

Again, $T(\alpha u_1) = T(\alpha(x_1, y_1)) = T(\alpha x_1, \alpha y_1)$

$= (3\alpha x_1 + \alpha y_1, \alpha x_1 + 2\alpha y_1)$

$= \alpha(3x_1 + y_1, x_1 + 2y_1)$

$\therefore T(\alpha u_1) = \alpha T(u_1)$

So, T is a linear mapping.

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4) Eigenvalues and Eigenvector

Question:

$A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$. Find the eigenvalues and eigenvectors of A and verify the Cayley-Hamilton Theorem for the matrix A.

Answer:

Given, $A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$ $\therefore I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

The characteristic matrix of A is, $\lambda I - A = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$

$\Rightarrow \lambda I - A = \begin{bmatrix} \lambda - 3 & 0 \\ -1 & \lambda - 2 \end{bmatrix}$

The characteristic polynomial of A is $|\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 \\ -1 & \lambda - 2 \end{vmatrix}$

$\Rightarrow |\lambda I - A| = (\lambda - 3)(\lambda - 2) + 0$

$\therefore$ The characteristic equation of A is $|\lambda I - A| = 0$

$\Rightarrow (\lambda - 3)(\lambda - 2) = 0$

$\therefore \lambda = 3$ and $\lambda = 2$

$\therefore \lambda = 3$ and $\lambda = 2$ are the eigenvalues of matrix A

We know that, $\vec{x}$ is an eigenvector of the matrix A corresponding to the eigenvalue λ if and only if,

$A\vec{x} = \lambda \vec{x}$

$\Rightarrow (\lambda I - A)\vec{x} = 0$ --- (1)

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Let, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be an eigenvector of the matrix A.

Then from ① we get,

$$(\lambda I - A)\vec{x} = 0$$

$$\Rightarrow \begin{bmatrix} \lambda-3 & 0 \\ -1 & \lambda-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad --- (2)$$

Now, if $\lambda = 3$, then, from equation (2) we get,

$$\begin{bmatrix} 3-3 & 0 \\ -1 & 3-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_2 = 0$$

$$\Rightarrow x_1 = x_2$$

If we choose $x_2 = 1$ then $x_1 = 1$

$\therefore$ Eigenvector, $\vec{x_1} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Again, if $\lambda = 2$, then from equation (2) we get,

$$\begin{bmatrix} 2-3 & 0 \\ -1 & 2-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$$\Rightarrow \begin{bmatrix} -1 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow -x_1 = 0 \quad \therefore x_1 = 0$

$-x_1 = 0 \quad \therefore x_1 = 0$

The variable x_2 is free. let, $x_2 = 1$.

$\therefore$ Eigenvector, $\vec{x_2} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

The Cayley-Hamilton Theorem states that every square matrix satisfies it's own characteristic equation.

Now, in order to verify Cayley-Hamilton theorem we need to show that, $(A-2)(A-3) = 0$

$\Rightarrow A^2 - 5A + 6I = 0$

$$\therefore A^2 = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3\times3+0\times1 & 3\times0+0\times2 \\ 1\times3+2\times1 & 1\times0+2\times2 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 \\ 5 & 4 \end{bmatrix}$$

$$\therefore A^2 - 5A + 6I = \begin{bmatrix} 9 & 0 \\ 5 & 4 \end{bmatrix} - \begin{bmatrix} 15 & 0 \\ 5 & 10 \end{bmatrix} + \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 9-15+6 & 0-0+0 \\ 5-5+0 & 4-10+6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore$ Cayley-Hamilton Theorem verified.

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5) Diagonalization of a Matrix and Inner Product Space.

Question:

i) Find the matrix P that diagonalize the matrix A and determine $D = P^{-1}AP$ ~~and~~, where D is the diagonal matrix. $A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$

ii) In R^2, find the norm of the vector $u = (4, -3)$ with the respect to the standard inner product

Answer:

i) Characteristic Equation of A is, $|\lambda I - A| = 0$

$$\Rightarrow \begin{vmatrix} \lambda - 3 & 0 \\ -1 & \lambda - 2 \end{vmatrix} = 0 \Rightarrow (\lambda - 3)(\lambda - 2) = 0$$

$\therefore \lambda = 3$ and $\lambda = 2$

Now, $(\lambda I - A)\vec{x} = 0$

$$\Rightarrow \begin{bmatrix} \lambda - 3 & 0 \\ -1 & \lambda - 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

For $\lambda = 3$, $x_1 - x_2 = 0 \Rightarrow x_1 = x_2$

$\therefore \vec{x_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda = 2$, $x_1 = 0$. Here, x_2 is free variable. Let, $x_2 = 1$

$\therefore \vec{x_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

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Now suppose that P is the matrix which has the above two eigenvectors as columns:

Then, $P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

Now, $|P| = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1 - 0 = 1 \neq 0$

$\therefore P^{-1} = \dfrac{Adj(P)}{|P|} = \dfrac{1}{|P|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$\therefore P^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

Now, $P^{-1}AP = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 3\times1+0\times1 & 3\times0+0\times1 \\ 1\times1+2\times1 & 1\times0+2\times1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 3 & 2 \end{bmatrix}$

$= \begin{bmatrix} 1\times3+0\times3 & 1\times0+0\times2 \\ -1\times3+1\times3 & -1\times0+1\times2 \end{bmatrix}$

$= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

$\therefore D = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

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ii)

$$\|u\| = \sqrt{x^2+y^2}$$

$$= \sqrt{4^2+(-3)^2}$$

$$= \sqrt{25}$$

$$= 5$$

$\therefore$ The norm of the vector u is 5